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REV STATUS OF SHEETS	REV									
	SHEET									

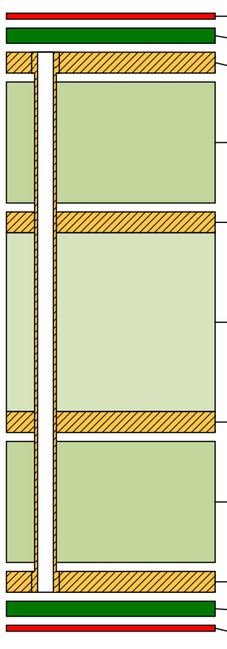
REVISIONS

ZONE	REV	DESCRIPTION	DATE	APPROVED

BEOVOX-POE v2.0

Stackup based on impedance controlled JLCPCB stackup JLC041611-7628

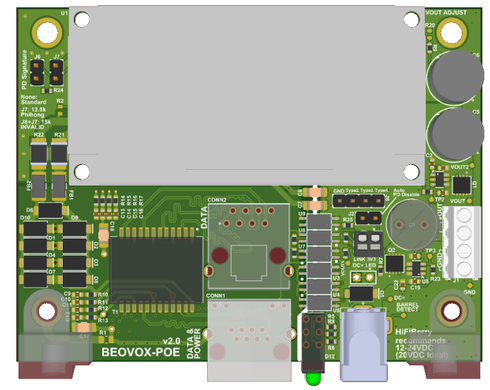
Layer Stack Legend



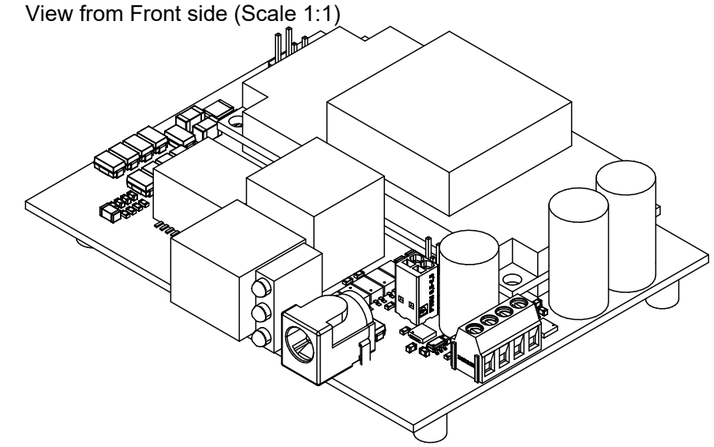
Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.03mm	Solder Resist	Solder Mask	GTS
Copper	**Top Layer**	**0.04mm**		**Signal**	**GTL**
Prepreg		*0.20mm*	*PP-006*	*Dielectric*	
CF-004	**Layer 2**	**0.04mm**		**Signal**	**G1**
Core		*1.03mm*	*FR-4*	*Dielectric*	
CF-004	**Layer 3**	**0.04mm**		**Signal**	**G2**
Prepreg		*0.20mm*	*PP-006*	*Dielectric*	
Copper	**Bottom Layer**	**0.04mm**		**Signal**	**GBL**
Surface Material	Bottom Solder	0.03mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

Total thickness: 1.63mm

Realistic View



View from Front side (Scale 1:1)



PCB Configuration

Copper Thickness	1oz, all layers
Soldermask	Black or Green
Silkscreen	White
Surface	HASL or ENIG
Minimum Clearance	6mils
Minimum Hole Size	0.3mm
Blind Vias	No
Impedance Match	Recommended
Material	FR-4
Via Fill	No
Board Thickness	1.6mm

Transmission Line Structure Table

Impedance Id	Transmission Line	Target Impedance	Calculated Impedance	Trace layer	Trace Width	Gap	Reference layers	Target Tolerance
1	Edge-Coupled Coated Microstrip	100	99.54	Top Layer	0.206mm	0.190mm	Layer 2	10%
2	Edge-Coupled Coated Microstrip	100	99.54	Bottom Layer	0.206mm	0.190mm	Layer 3	10%

PART NO: =PCB_PART_NUMBER

APPROVALS		DATE
ENGINEER:	JOSH HORNE	JOSH
DESIGNER:	=PCB_DESIGNER	=PCB_DESIGNER
CHECKER:	=PCB_CHECKER	=PCB_CHECKER

Reference Documents

BOM DOC:	=DOC_NO_BOM
ASSY DOC:	=DOC_NO_FAB_DWG
SCH DOC:	=DOC_NO_SCH_DWG
PCB DOC:	=PCB_DWG_NO

THIRD ANGLE PROJECTION

NEXT ASSY | USED ON

APPLICATION

Altium

ADDRESS 1
ADDRESS 2
ADDRESS 3
ADDRESS 4

DESIGN ITEM: .Item
DESIGN ITEM REVISION: v2.0

TITLE: BEOVOX-POE v2.0

SIZE: B | CAGE CODE: =CAGE_CODE | DWG NO: | REV:

SCALE: | FILE NAME: beovox-poe.PCBDwf | SHEET: 1 OF 2

DWG NO: =DOC_NO_ASSY_DWG REV: v2.0

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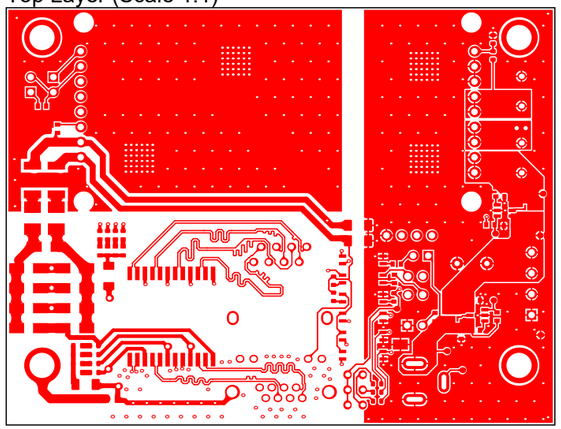
DWG NO: =DOC_NO_ASSY_DWG REV: v2.0

REV STATUS OF SHEETS	REV									
	SHEET									

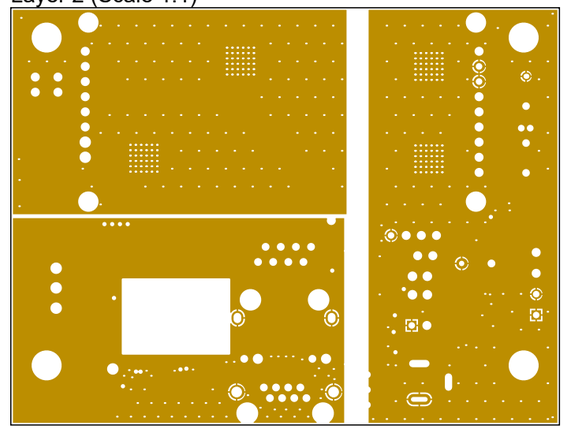
REVISIONS

ZONE	REV	DESCRIPTION	DATE	APPROVED

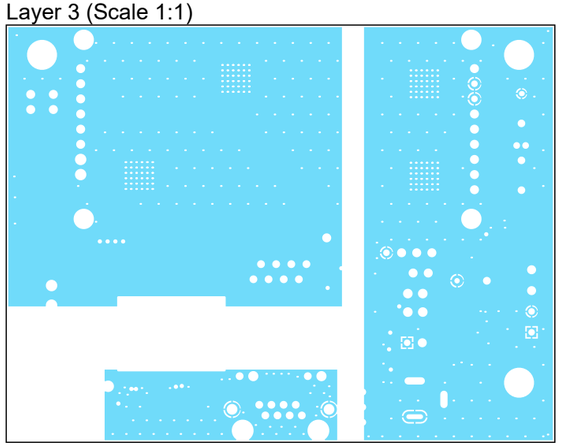
Top Layer (Scale 1:1)



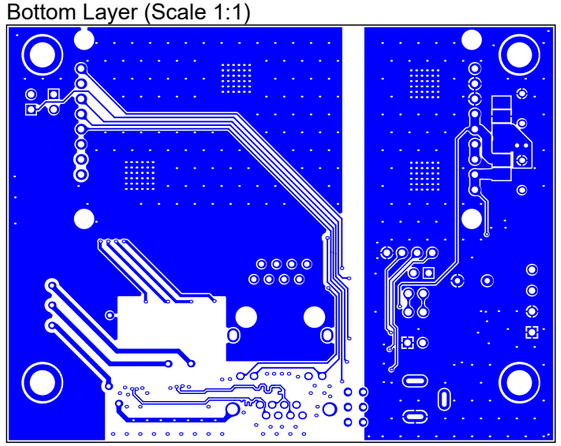
Layer 2 (Scale 1:1)



Layer 3 (Scale 1:1)



Bottom Layer (Scale 1:1)



PART NO: =PCB_PART_NUMBER

APPROVALS		DATE
ENGINEER:	JOSH HORNE	JOSH
DESIGNER:	=PCB_DESIGNER	=PCB_DESIGNER
CHECKER:	=PCB_CHECKER	=PCB_CHECKER

Reference Documents

BOM DOC:	=DOC_NO_BOM
ASSY DOC:	=DOC_NO_FAB_DWG
SCH DOC:	=DOC_NO_SCH_DWG
PCB DOC:	=PCB_DWG_NO

THIRD ANGLE PROJECTION

NEXT ASSY | USED ON

APPLICATION

Altium™

ADDRESS 1
ADDRESS 2
ADDRESS 3
ADDRESS 4

DESIGN ITEM: .Item

DESIGN ITEM REVISION: v2.0

TITLE: BEOVOX-POE v2.0

SIZE: B | CAGE CODE: =CAGE_CODE | DWG NO: | REV:

SCALE: | FILE NAME: beovox-poe.PCBDwf | SHEET: 2 OF 2

DWG NO: =DOC_NO_ASSY_DWG REV: v2.0